

Simple Banking System

Assignment Project Documentation

Project Overview

This project is assigned about the Simple Banking System against an interview session. Within the assigned time duration I tried to complete the project. This project has two parts are following:

- 1) Backend
- 2) Frontend

Features developed

- 1) Role based user management.
- 2) Customer management.
- 3) Account Management.
- 4) Transaction management.
- 5) Separate "DEPOSIT", "WITHDRAWAL", "ACCOUNT_TRANSFER" operations developed.
- 6) Searching transaction history.
- 7) Download Account statements
- 8) Debit/Credit color Indicator.
- 9) LoggedIn user profile view.
- 10) Unique Account Number generation.
- 11) Validation messages.

Technologies

I used several technologies in this project to develop.

Backend: Java, Spring Boot-4.0.0, Maven.

Frontend: Angular 19.2.19 , NodeJS 18

Database: Oracle Database

Backend project setup

Please clone the project from the repository:

<https://github.com/gitsayed/simple-banking-system>

- 1) Now check that your oracle database is installed in your machine and is running.

- 2) Create a database with your desired name.
- 3) Go to the project “backend” folder and it is the backend project.
- 4) There you find a file **applications.properties** file in this location
/backend/src/main/resources/applications.properties
- 5) Change the following properties with database information:
 - a) **spring.datasource.url=jdbc:oracle:thin:@localhost:1521/TEST_BANK**
 - b) **spring.datasource.username=appuser**
 - c) **spring.datasource.password=appuser123**
- 6) Set your “**server.port=**” property with your port.
- 7) Please execute SQL commands in your database which stored in the following location **/backend/src/main/resources/DB_FILES/InitialScript.sql**
- 8) Now you need JDK and Maven in your machine. I used JDK 25 in this backend project.
- 9) When installed JDK and Maven, later run the following commands one after one:
 - a) **mvn clean package -DskipTests**
 - b) **mvn spring-boot:run**
- 9) Now you will see that the project will start with the desired port and machine IP Address.
- 10) Now you can test the APIs from these url:
<http://localhost:8080/cbs/swagger-ui/index.html>

Frontend project setup

Please clone the project from the repository:

<https://github.com/gitsayed/simple-banking-system>

- 1) Please see the frontend project in the cloned location frontend folder.
- 2) I used Node Js 18 and Angular 19.2.19 to develop the frontend project. So, please install those in your machine.
- 3) Please configure [environment.ts](#) file file to connect your backend project with frontend project.
- 4) Now run following commands:
 - a) “npm install”
 - b) “ng serve”
- 5) The Angular project opens on <http://localhost:4200> port by default. You can test the project right now.
- 6) A default user stored in the SQL file in the backend project. Which
 - a) Username: “admin”,
 - b) Password: “admin”,
 - c) Roles: [ADMIN]

Conclusion:

This is my assigned project. The duration of the project development is five days. I tried to develop the project with my highest efforts. There may be some mistakes after this. I apologize for that.

Thank you all.